Nuclear Fuel Performance

NE-533

Spring 2026

Housekeeping

- Exam 3 stats:
  - Avg: 84
  - Curve of 6 points applied
- Presentation and MOOSE part 2 grades out this week

- Plan is for the last exam to be on April 23 (week from today)
- Last partial lecture and problem session next Tuesday
- Reminder MOOSE part 3 is due April 28

- ClassEval is active, please everyone fill this out
- Send me the confirmation that you completed this, and you get +3 onto exam 4
- Will also send out a link to my own personal survey (this is also anonymous), send me the confirmation you completed this, get another +3 onto exam 4

Last Time

- Steam kinetics
  - Beyond 800 C oxidation is too rapid, above 1200 C, entire cladding gets consumed
  - Zr phase transformation at 863 C, leads to beta and O-stabilized alpha (which is more brittle)
- ATF
  - improved cladding properties, improved steam corrosion, improved fuel properties, improved fission product retention
  - Near term improvements: coatings, fuel additives
  - Novel fuel/cladding: U3Si2/UN fuels, FeCrAl cladding
  - Transformative fuel/cladding: FCM, SiC/SiC, etc.
- Limiting phenomena: Pellet-clad mechanical interaction; Cladding elongation and assembly bow; Cladding oxidation and hydrogen pickup; Cladding wear; Power to melt; Fuel rod internal pressure; Departure from nucleate boiling; Normal operation limits

WATER CHEMISTRY

Water Chemistry

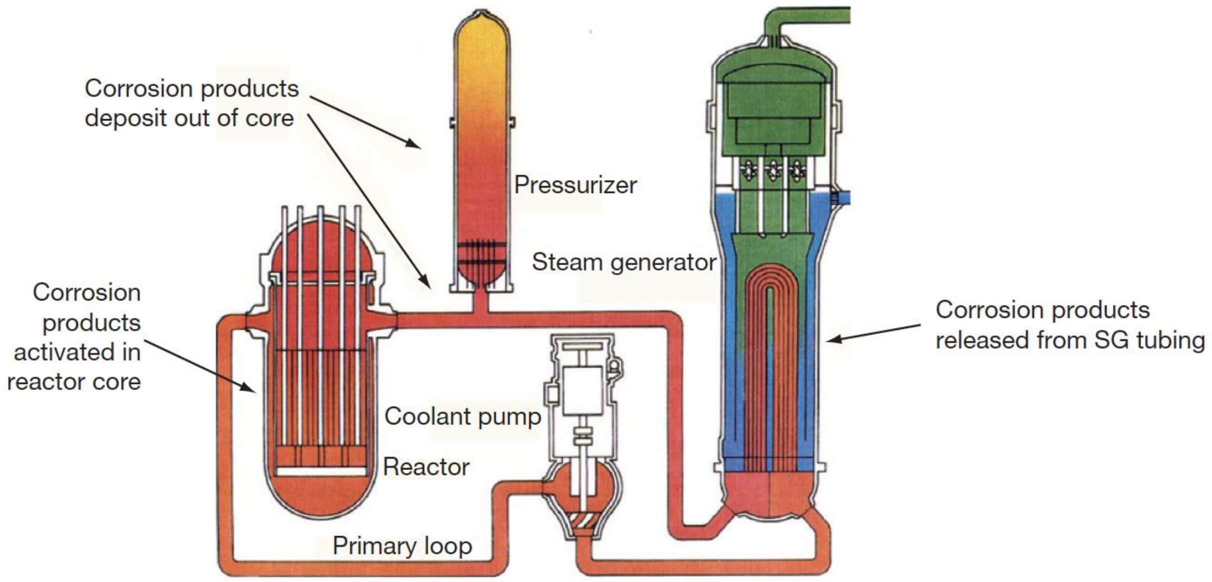
- Excellent water quality is essential if material degradation is to be controlled
- Primary system water chemistry affects fuel performance through the deposition of corrosion products on fuel pin surfaces
- In the early days of nuclear power plant operation, impurities in the coolant water were a major factor in causing excessive corrosion
- Chlorides and sulfates are particularly aggressive in increasing intergranular stress corrosion cracking (IGSCC) and other corrosion processes
- Initial efforts to improve water quality brought about a slow but steady reduction in impurities through improved design and operation of purification systems

Water Chemistry

- Excellent water chemistry alone is not sufficient to control corrosion, thus programs to modify water chemistry, including minimizing oxygen to reduce the electrochemical corrosion potential (ECP) in BWRs, and oxygen and pH control in PWRs, have been implemented
- Additives to further inhibit the corrosion process have been developed and are now in widespread use
- Water chemistry advances are now an important part of the overall operating strategy to control material degradation

PWR Water Chemistry

- In the very early days of PWR operation, heavy crud buildup on fuel cladding surfaces was caused by the transport of corrosion products from the steam generators into the reactor core
- Activated corrosion products caused high-radiation fields on out-of-core surfaces
- Fuel performance was compromised, and even coolant flow issues were observed



CRUD

- A corrosion product called Chalk River unidentified deposit (CRUD) accumulates on the Ni alloy and stainless-steel surfaces
- CRUD is an accumulation of materials and corrosion products that is composed of either dissolved ions or solid particles such as Ni, Fe, and Co on fuel rod cladding surfaces
- CRUD degrades heat production by nuclear fuel because the constituents are slowly eroded by the circulation of the hot pressurized water and later deposited on the cladding
- The chemical composition of CRUD varies depending on the types of refueling cycles and the constituents of the basic metal material

CRUD Cycle

- First, metallic constituents from core internals or heat exchangers are dissolved in the high T water
- Those constituents get deposited on the surface of the cladding, often in a spinel-type structure (AB2O4), with Fe, Ni, and Co some of the primary species
- Proximity to the fuel leads to activation of these species, producing radionuclides in the CRUD, such as 60Co and 63Ni
- CRUD can flake off, chip, or species can dissolve into the water and then collect on the out-of-core surfaces

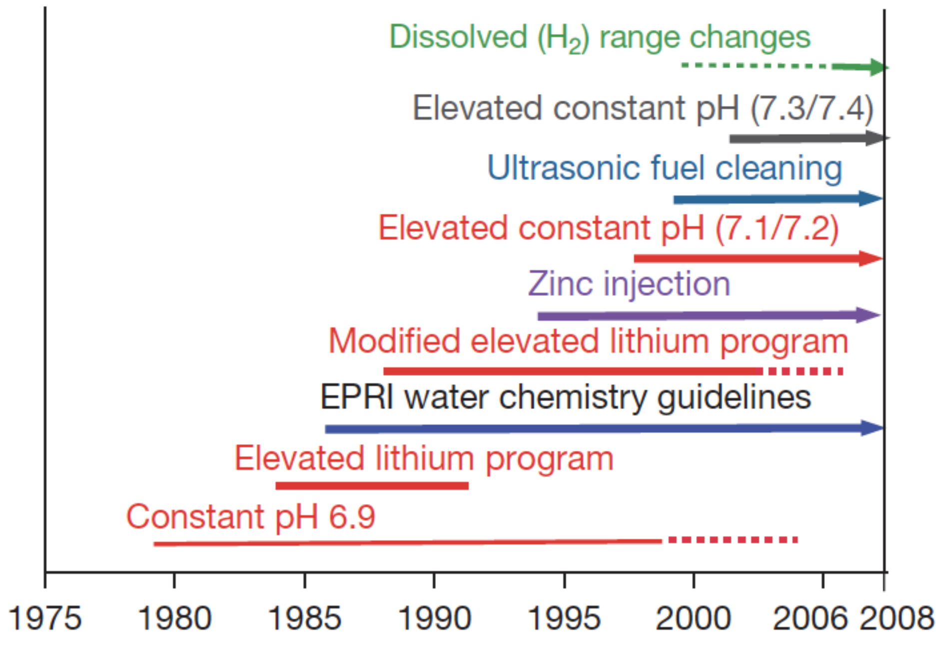
PWR Water Chemistry

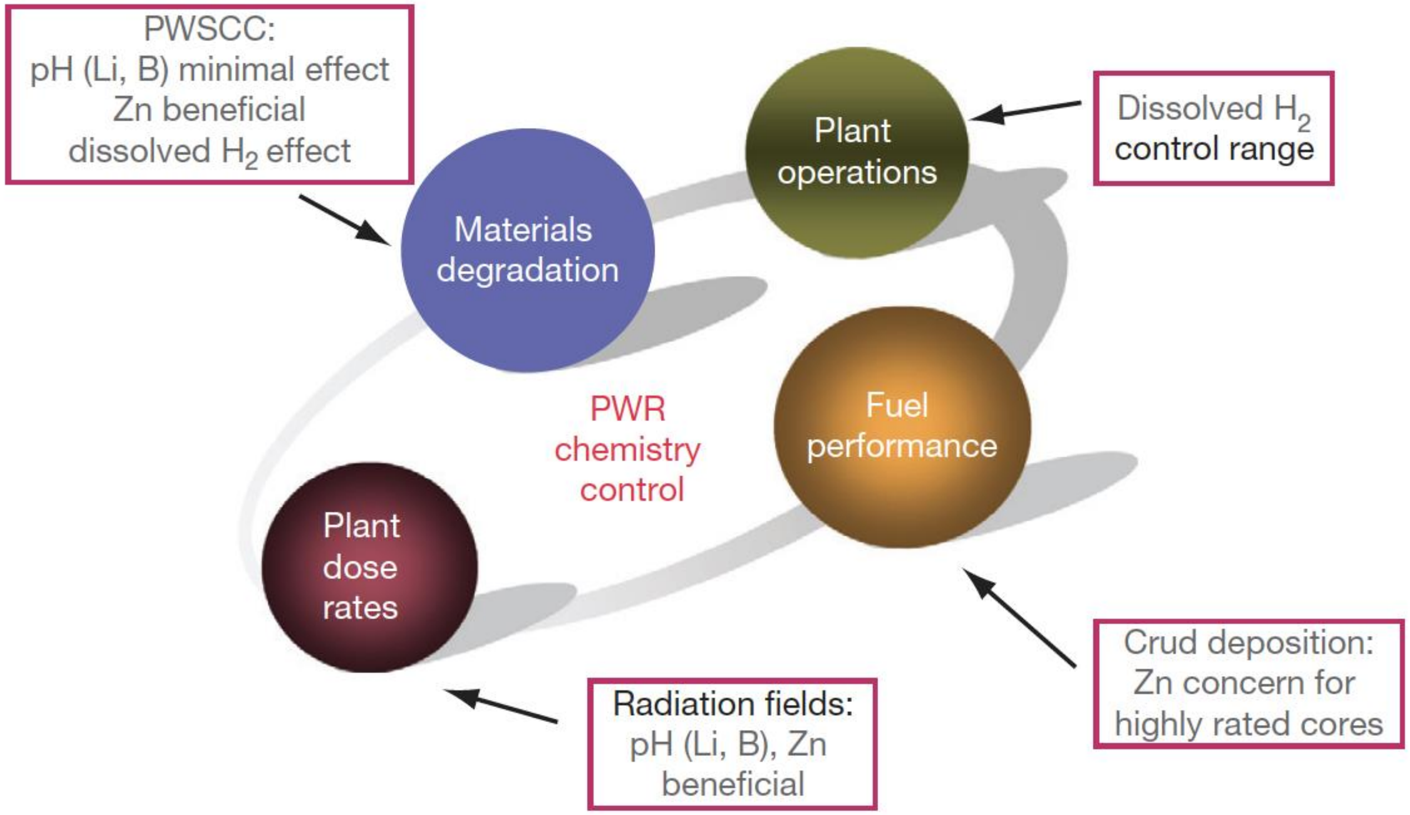
- PWR problems were initially mitigated by imposing a hydrogen overpressure on the primary system, reducing the corrosion/oxygen potential, and raising the primary chemistry pH
- Commercial PWR power plants use a steadily decreasing concentration of boric acid as a chemical shim for reactivity control throughout the fuel cycle, which results in the use of lithium hydroxide to control pH
- The concept of ‘coordinated boron and lithium’ was developed, whereby the concentration of LiOH was gradually reduced in line with the boric acid reduction to maintain a constant pH
- It was determined that heavy fuel crud buildup was avoided if a constant pH of > 6.9 was maintained, as the solubility of Ni and Fe is dependent on pH
- Zinc injection is utilized to reduce radiation fields, and also inhibits SCC

Radiation Control

- Corrosion products deposited on the fuel become activated, are released back into the coolant, and may be deposited on out-of-core surfaces
- During shutdowns, the major radiation source for personnel exposure is activated corrosion products, deposited on primary system surfaces
- The mechanism of the zinc ion effect is complex, as release of ^{60}Co from fuel crud is reduced, and deposition out-core is also reduced
- Aqueous zinc ion promotes the formation of a more protective spinel-structured corrosion film on stainless steel, especially when reducing conditions are present
- Both cobalt and zinc favor tetrahedral sites in the spinel structure, but the site preference energy favors zinc incorporation
- The ^{60}Co remains longer in the water and is eventually removed by the cleanup system

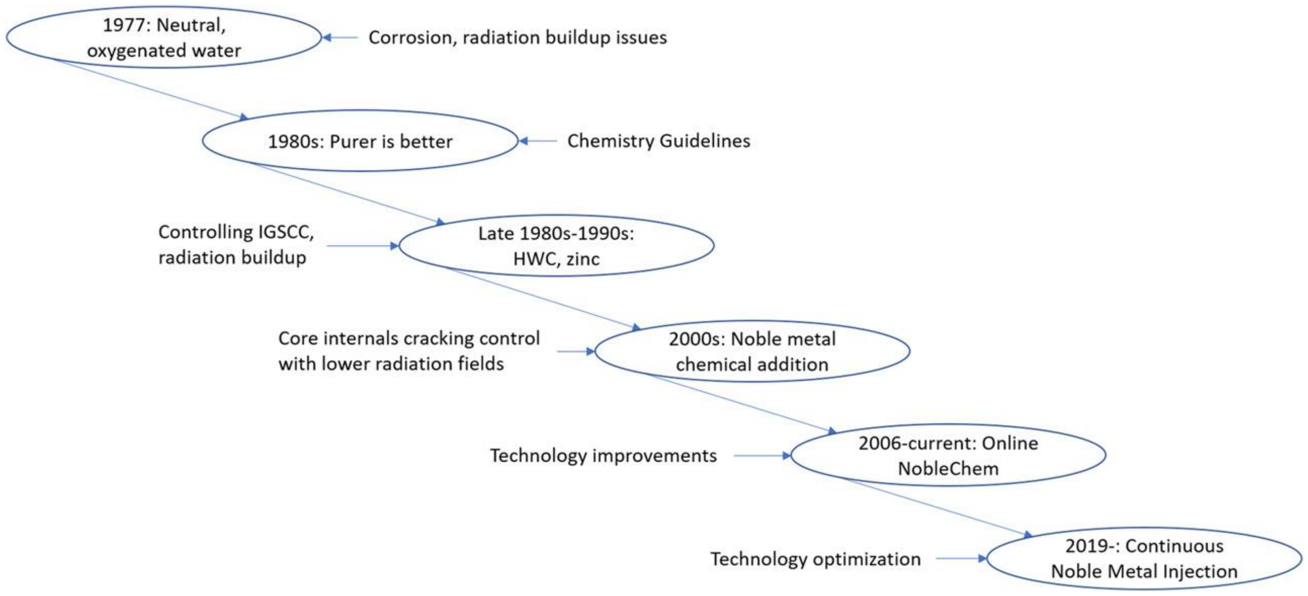
PWR Water Chemistry

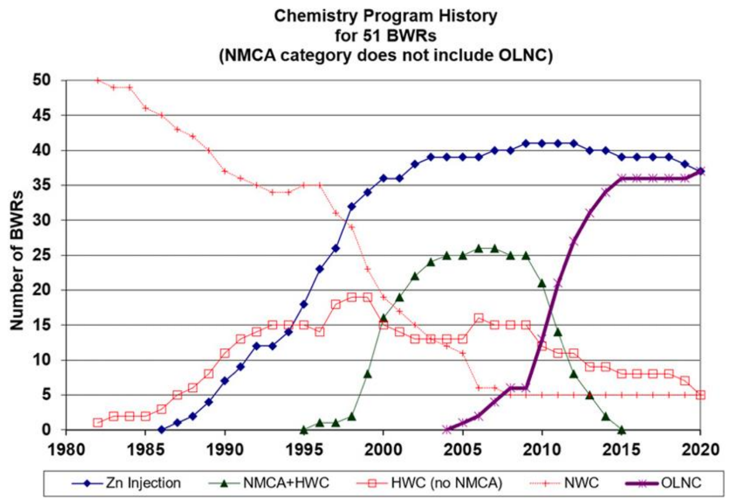




BWR Water Chemistry

- Similarly, BWR water chemistry has to be optimized to meet requirements on material degradation, fuel performance, and control of radiation fields
- BWR chemistry strategies have changed over time, focusing on purity, limiting SCC, and control of radiation fields





IGSCC Mitigation

- Intergranular SCC (IGSCC) of 304 stainless steel core internals (and other materials) is one of the key chemistry control issues in BWRs
- The control of the electrochemical potential (Redox potential) by hydrogen injection was effective at reducing IGSCC
- Move from normal water chemistry (NWC) to hydrogen water chemistry (HWC) limits crack growth
- Added noble metals (such as Pt) as coating on surfaces, or injection into water to increase efficiency of HWC

IGSCC Mitigation

- Noble metal addition had previously only been conducted during outages, but changed to online addition to prevent ‘crack flanking’
- Areas in the core are protected by different mitigation strategies, based on temperatures and two-phase flow

- Radiation control: Zn injection also used in BWRs

Fuel Performance Concerns

- CRUD deposition on the cladding surface can reduce heat transfer, increasing fuel temperatures, which also increases oxidation rates
- Addition of Zn and noble metals can increase the adherence of CRUD deposits on the cladding
- Zn/noble metals also can cause a failure/cracking of the oxide layer, promoting further corrosion
- There are concentration limits on Zn and noble metals to limit CRUD formation
- In PWRs, CRUD-induced power shifts (CRIPs/CIPS) can occur by trapping boron in the CRUD, changing the power distribution
- CRUD-related failures have been eliminated since about 2005

Key Themes Over Time

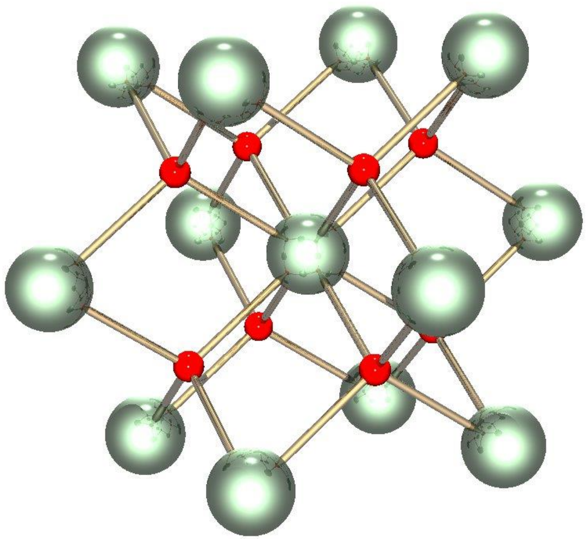
- Increasing Chemical Precision
  - From simple purity control to tightly optimized ppm/ppb-level adjustments
- Shift from Reactive to Predictive
  - Early chemistry responded to problems; modern chemistry anticipates them via modeling
- Tradeoff Management
  - Every chemistry choice affects multiple outcomes:
    - Higher pH leads to less corrosion, but possible fuel issues
    - Hydrogen injection → less SCC, but higher dose rates (BWRs)
- Materials–Chemistry Coupling
  - Water chemistry evolution closely tracks:
    - Alloy development (e.g., Inconel, stainless steels, Zircaloy)
    - Fuel design changes (higher burnup, different claddings)

Summary

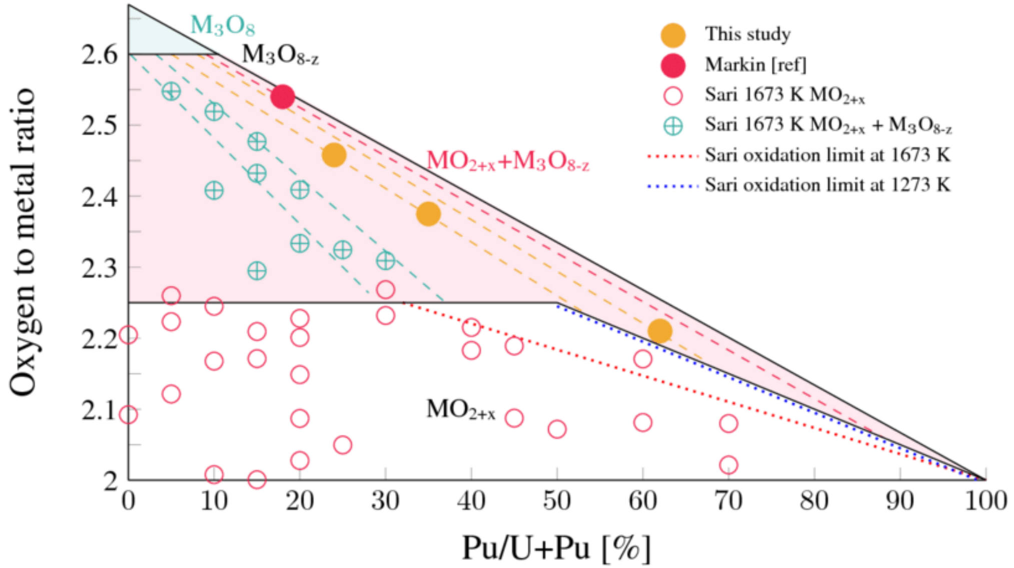
- Primary system water chemistry affects fuel performance through the deposition of corrosion products on fuel pin surfaces
- Control measures such as dissolved H2 and balancing LiOH to boron content are utilized to control the pH
- Zinc injection is utilized to control radiation fields
- CRUD can cause failures or CRIPs, affecting the performance of the fuel
- Water chemistry is an evolving balancing act between corrosion of the core internals, radiation fields, CRUD development, pH, and reactivity

MIXED OXIDE (MOX) FUELS

Why MOX?



- The first fast breeder reactors built in the 1950s used metallic fuel (plutonium and uranium), as metals offer the highest heavy metal density and therefore the highest breeding ratio
- Because of dimensional instability due to swelling and growth, metal fuels couldn't achieve high burnup
- By the 1960s, mixed uranium and plutonium oxide (U,Pu)O2=MOX was known to be highly radiation tolerant and began to be considered as a reference fuel for fast reactors



Mixed Oxides

- MOX fuel allows to burn excess weapons grade plutonium
- About 30 LWRs in Europe currently utilize a partial MOX core
- Similar behavior to UO2, but different neutronics, fission gas release, thermal conductivity, etc.
- Less common is inclusion of minor actinides in MOX to burn waste
- Can also serve as a breeder
- Also a fuel concept for SFRs

Table 1 Main characteristics of standard fuel pins irradiated in the prototype and commercial fast reactors ($p > 200$ MWth)

	BN350	*Phénix*	*PFR*	*BN600*	*FFTF*[a]	*Super-Phénix*	*MONJU*
First criticality	1972	1973	1974	1979	1980	1985	1994
Thermal power (MWth)	750	563	600	1470	400	3000	714
Electric power (MWe)	350[b]	250	250	600	–	1200	280
Type of fuel	UO_2	$(U,Pu)O_2$	$(U,Pu)O_2$	UO_2	$(U,Pu)O_2$	$(U,Pu)O_2$	$(U,Pu)O_2$
No. of subassemblies (inner/outer core)	109/117	55/48	28/44	209/160	28/45	193/171	108/90
No. of pins per assembly	127	217	325	127	217	271	169
Type of spacer	Wire	Wire	Grids	Wire	Wire	Wire	Wire
Length of pin (m)	1.8	1.793	2.25	2.445	2.38	2.7	2.813
Height of fissile column (m)	1.06	0.85	0.914	1.0	0.914	1.0	0.93
Lower fertile column length (m)	0.4	0.3	0.45	0.4	–	0.3	0.35
Upper fertile column length (m)	0.57	0.31	0.45	0.4	–	0.3	0.3
Clad outer diameter (mm)	6.9	6.55	5.8	6.9	5.84	8.5	6.5
Clad thickness (mm)	0.4	0.45	0.38	0.4	0.38	0.565	0.47
Helical wire diameter (mm)		1.15			1.42	1.2	1.32
Pellet diameter (mm)		5.42				7.14	5.4
Fuel clad diametral gap (mm)		0.23			0.14	0.23	0.16
Central hole diameter (mm)	0	0	1.5	0		2.0	0
Fissile atoms/(U + Pu) (%) (inner core/outer core)	17/26	18/23	22/28	17/26	20/25	15/22	16/21
Fuel density (% TD)	95	95.5	97	95	91	95.5	85
Smeared density (%)	75	88	78	77	86	83	80
Plenum volume (cm^3)	8	13	14	21	19	43	28
Maximum linear power ($W\,cm^{-1}$)	400	450	420	472	413	470	360
Peak cladding temperature (°C)	570	650	670	700	660	620	675
Maximum neutron flux ($10^{15}\,n\,cm^2\,s^{-1}$)	7	7.1	7.6	7.7	7	6	6.0
Maximum burnup (at.%)	9.0	16.9	23.5	11.8	24.5	Not relevant	Not relevant
($GWd\,t^{-1}$)	65	144	200				
Maximum dose (dpa)	60	156	155	90		–	–

MOX Designs

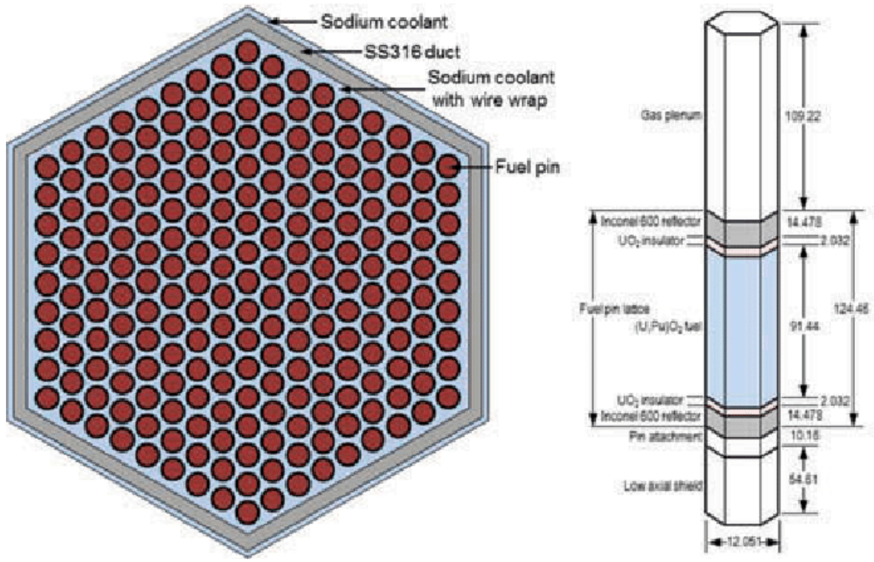
- The fuel pin is a long cylinder (2–3m long, 5–10mm diameter), clad in a steel tube (0.4–0.6mm thick) closed in both ends by welded plugs, preventing direct contact between the radioactive material and the sodium coolant (or water coolant, in LWR concepts)
- The oxide fissile column (~1 m long) consists of a stack of conventionally pressed and sintered pellets with an outer diameter slightly smaller than the inner diameter of the clad, providing a gap ~100 micron gap
- Both full pellets and annular pellets have been used
- A He gap with a pressure ~1 atm is used
- Fuel pins of fast reactors are designed to operate at a high linear heat generation rate: between 400 and 500 W/cm, about 2x-3x higher than standard linear power in light water reactors (LWRs)

MOX Fuel

- As the fast reactor fuel pin diameters are generally smaller than classical rod diameters of LWRs, the power density and heat fluxes are much higher in SFRs than in LWRs
- For example, in a Phenix fuel pin at 450 W/cm, the power density in the pellet reaches almost 2000 W/cc
- Sodium enters the bottom part of the core at about 400 C and the average coolant temperature above the core is typically about 550 C
- The neutron flux is very intense (~7×10^{15} n/cm^2/s in the core center) and the assembly materials suffer high damage, more than 100 dpa, at high burnup
- Qualifying metallic materials able to withstand such high damage while keeping their shape and mechanical properties was/is one of the key challenges for fast reactors
- In order to reduce fuel cycle costs, the main objective of oxide fuels R&D for fast reactors has been to reach high burnup, typically around 150 GWd/ton, about twice the burnup achieved in LWRs

UO2 to MOX Design Differences

- Shorter fuel column/rod in MOX
- Smaller fuel diameter in MOX
- Steel cladding vs. zircaloy cladding
- Much higher heating rates in MOX
- Larger gas plenum
- Sodium coolant in SFRs
- Extended burnup for MOX
- Hexagonal packing of fuel pins



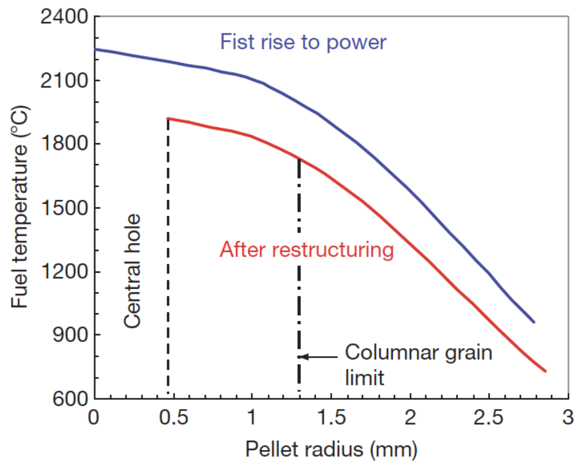
FFTF Assembly

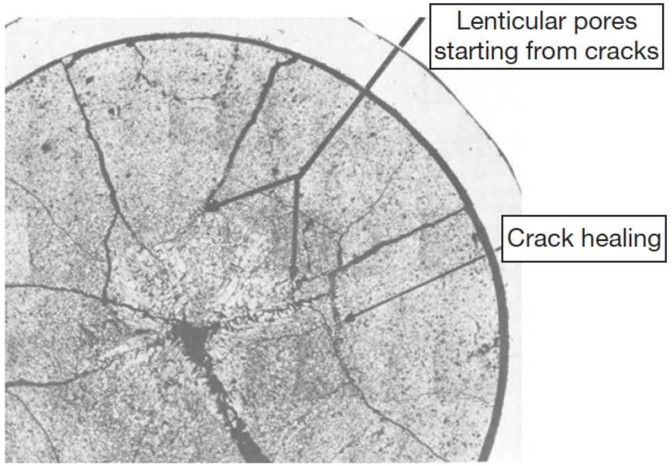
MOX Fuel

- The main requirements and design criteria are the following:
  - Guaranteeing the absence of fuel melting, both in nominal conditions and during off-normal events
  - Keeping cladding integrity and fuel pin tightness
  - Cooling of the fuel pin bundle must be ensured up to high burnup in all operating conditions
  - Loading and unloading of subassemblies have to be guaranteed, which induces a limitation on the deformation of the hexagonal wrapper tubes

MOX Fuels

- Most phenomena occurring inside oxide fuel pellets are thermally activated, and a good knowledge of the thermal field inside the fuel stack is key
- Centerline temperatures can reach above 2000C, significantly higher than thermal LWRs
- Thermal conductivity is low and degrades with irradiation
- Due to the steep thermal gradient, thermal stress cracks form
- Restructuring takes place due to the high temperatures, leading to distinct regions in the fuel





MOX Restructuring

- Pu bearing fast reactor oxide fuels display four defining regions of a restructured pellet:
  - the central void, the columnar grain growth region, the equiaxed grain growth region, and the as-sintered region